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Vertical Sum

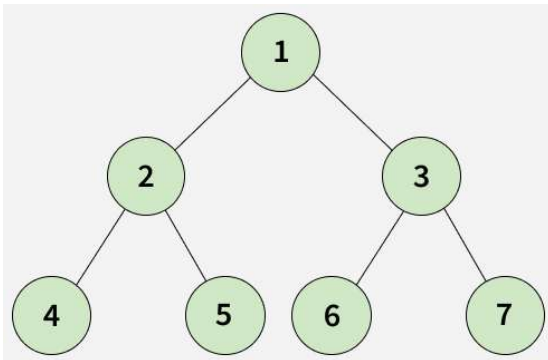
Difficulty: **Medium**Accuracy: **64.76%**Submissions: **61K+**Points: **4**Average Time: **20m**

Given a binary tree having n nodes, find the vertical sum of the nodes that are in the same vertical line. Return all sums through different vertical lines starting from the left-most vertical line to the right-most vertical line.



Examples:

Input:

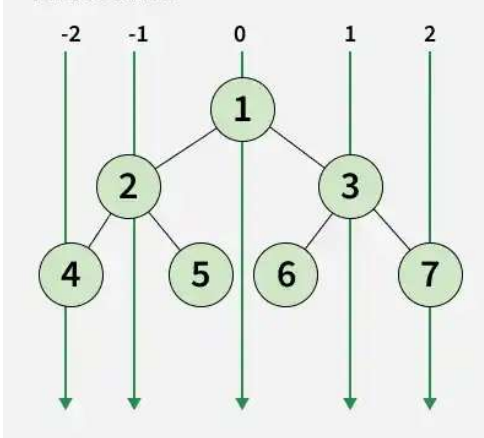


Output:

4 2 12 3 7

Explanation:

Vertical Lines:



The tree has 5 vertical lines

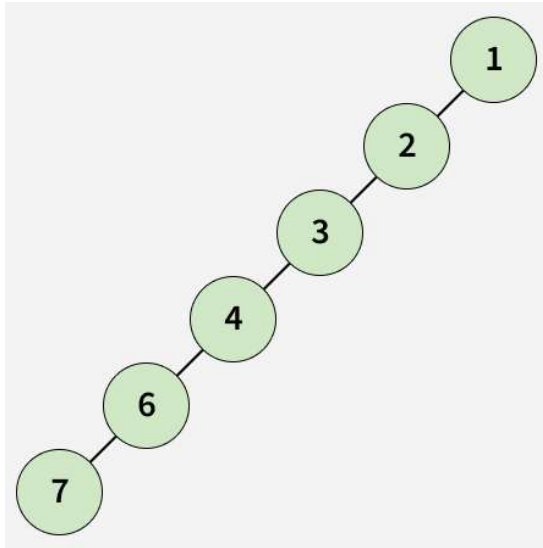
Line 1 has only one node 4 => vertical sum is 4.

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Line-4 has only one node 3 => vertical sum is 3.

Line-5 has only one node 7 => vertical sum is 7.

Input:**Output:**

7 6 4 3 2 1

Explanation:

There are six vertical lines each having one node.

Constraints:

$$1 \leq n \leq 10^4$$

$$1 \leq \text{Node value} \leq 10^5$$

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Python3



```
1 from collections import defaultdict, deque
2
3 class Solution:
4     def verticalSum(self, root):
5         if not root:
6             return []
7
8         mp = defaultdict(int)
9         q = deque([(root, 0)]) # (node, horizontal distance)
10
11        while q:
12            node, hd = q.popleft()
13
14            mp[hd] += node.data
15
16            if node.left:
17                q.append((node.left, hd - 1))
18
19            if node.right:
20                q.append((node.right, hd + 1))
21
22        return [mp[hd] for hd in sorted(mp)]
```

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